PROMPT LOG – LAB SESSION 04

Subject: 23CSE301

Lab Session: 04

GenAI Tool Used: ChatGPT

Prompt 1 – Lab 03 Experiments

User Prompt:

I have Lab Session 04 for subject 23CSE301. I need to repeat all the experiments from Lab Session 03 using AI tools. The experiments include feature type identification, Label Encoding, One-Hot Encoding, Minkowski distance, dot product, Euclidean norm, mean, variance, standard deviation, histogram and k-means. Please provide modular Python code suitable for a Jupyter/Colab notebook.

AI Response Summary:

Provided modular Python functions and code for implementing the experiments from A1 to A11, including categorical encoding, Minkowski distance, vector operations, statistical functions, histogram generation and k-means clustering.

Prompt 2 – Feature Encoding

User Prompt:

Give me Python functions for Label Encoding and One-Hot Encoding for the marketing_campaign dataset. Keep the functions modular and don't put print statements inside the functions.

AI Response Summary:

Provided separate label_encode() and one_hot_encode() functions and example code for applying them to categorical columns.

Prompt 3 – Minkowski Distance

User Prompt:

Write a Python function to calculate generalized Minkowski distance between two vectors. It should support different values of p, including Manhattan distance when p=1 and Euclidean distance when p=2.

AI Response Summary:

Provided a modular minkowski_distance() function and example code for calculating distances for p values from 1 to 10.

Prompt 4 – Distance Comparison

User Prompt:

Give code to compare my own Minkowski distance function with scipy.spatial.distance.minkowski() for p from 1 to 10.

AI Response Summary:

Provided code comparing the custom implementation with SciPy and calculating the difference between the two results.

Prompt 5 – Dot Product and Norm

User Prompt:

Give Python functions for calculating dot product and Euclidean norm without directly using numpy.dot() and numpy.linalg.norm(). Then provide code to compare the results with NumPy.

AI Response Summary:

Provided custom functions for dot product and Euclidean norm and comparison code using NumPy functions.

Prompt 6 – Statistics

User Prompt:

Give modular Python functions to calculate mean, variance and standard deviation. Then calculate these values for every column of a numerical dataset.

AI Response Summary:

Provided separate functions for mean, variance and standard deviation and a function for calculating column-wise statistics.

Prompt 7 – NumPy Comparison

User Prompt:

Give code to compare the mean, variance and standard deviation calculated using my own functions with numpy.mean(), numpy.var() and numpy.std().

AI Response Summary:

Provided comparison code using np.allclose() to check whether the custom and NumPy results match.

Prompt 8 – Histogram

User Prompt:

Give code to select one numerical feature from the marketing_campaign dataset, calculate its mean and variance, and plot its histogram using buckets.

AI Response Summary:

Provided Python code using Matplotlib to generate a histogram and custom functions to calculate the mean and variance.

Prompt 9 – K-Means

User Prompt:

Implement the basic k-means clustering algorithm in Python using modular functions. Use the previously developed Minkowski distance function. The algorithm should select initial centroids, assign points to clusters, recompute centroids and stop when the centroids no longer change.

AI Response Summary:

Provided modular functions for assigning clusters, updating centroids and implementing the complete k-means algorithm.

Prompt 10 – Unit Testing

User Prompt:

Lab 04 requires unit test cases for the functions developed in Lab 03 and Lab 04. Generate Python unittest test cases for Minkowski distance, mean, variance, standard deviation, dot product and Euclidean norm.

AI Response Summary:

Provided a unittest.TestCase class containing multiple test cases for the modular functions.

Prompt 11 – Unit Test Error

User Prompt:

I am getting the error NameError: name 'unittest' is not defined while running my unit test code. How can I fix it?

AI Response Summary:

The error occurs because the unittest module was not imported. Add import unittest before defining the test class and run the test cell again.

Prompt 12 – K-Means Performance Comparison

User Prompt:

Lab 04 requires a performance comparison between my own k-means implementation and an AI-generated k-means implementation. Give me two versions of k-means and code to measure their execution time.

AI Response Summary:

Provided two k-means implementations and a performance measurement function using time.perf_counter(). The average execution time was calculated over multiple repetitions.

Prompt 13 – Performance Graph

User Prompt:

Give Python code to plot a bar graph comparing the execution time of the two k-means implementations.

AI Response Summary:

Provided Matplotlib code to create a bar chart showing the execution time of Version 1 and Version 2.

Prompt 14 – Final Lab 04 Code

User Prompt:

Give me the complete A1 to A11 code for Lab 04 in the correct order, including unit tests and k-means performance comparison.

AI Response Summary:

Provided the complete Jupyter/Colab implementation from A1 to A11, followed by unit testing and performance comparison of two k-means implementations.

Prompt Log Summary

ChatGPT was used as the GenAI tool for code generation, debugging, unit-test generation and performance-comparison support. The prompts were focused on implementing the experiments specified in Lab Session 03 and repeating them for Lab Session 04. The generated code was reviewed and adapted for the requirements of the assignment.